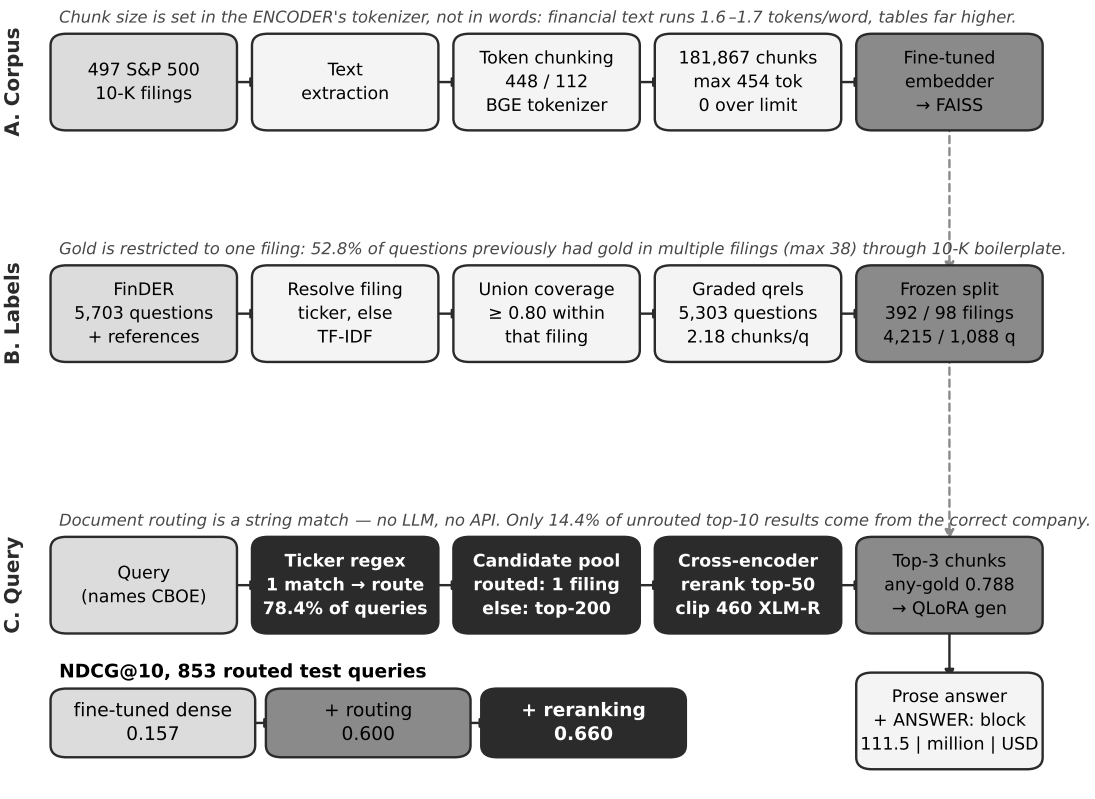

A. Corpus
Chunk size is set in the ENCODER's tokenizer, not in words: financial text runs 1.6–1.7 tokens/word, tables far higher.
497 S&P 500 10-K filings
Text extraction
Token chunking 448 / 112 BGE tokenizer
181,867 chunks max 454 tok 0 over limit
Fine-tuned embedder → FAISS
B. Labels
Gold is restricted to one filing: 52.8% of questions previously had gold in multiple filings (max 38) through 10-K boilerplate.
FinDER 5,703 questions + references
Resolve filing ticker, else TF-IDF
Union coverage ≥ 0.80 within that filing
Graded qrels 5,303 questions 2.18 chunks/q
Frozen split 392 / 98 filings 4,215 / 1,088 q
C. Query
Document routing is a string match — no LLM, no API. Only 14.4% of unrouted top-10 results come from the correct company.
Query (names CBOE)
Ticker regex 1 match → route 78.4% of queries
Candidate pool routed: 1 filing else: top-200
Cross-encoder rerank top-50 clip 460 XLM-R
Top-3 chunks any-gold 0.788 → QLoRA gen
Prose answer + ANSWER: block 111.5 | million | USD
NDCG@10, 853 routed test queries
fine-tuned dense 0.157
+ routing 0.600
+ reranking 0.660